

How to Link the Static Library (.lib) to a Keil MDK 5 Project

The LCD panel static graphic libraries (.lib) are located within the **KVB_LCD** project folder, alongside the **GUI** and evaluation files (.c and .h).

File Structure Overview

- **KVB_LCD.h** – Main **GUI** (Graphical User Interface) header file.
- **KVB_TEST.h** – Header file containing two evaluation/presentation functions tailored for different screen resolutions.
- **KVB_TEST.c** – Source file implementing the two resolution-dependent evaluation functions.
- **KVB_TEXT.h** – Header file dedicated to the text variables used in the evaluation code.
- **KVB_TEXT.c** – Source file containing the text variable definitions for the evaluation code.
- **LCD_16bit_Open407_F407VGT6_9325.lib** – Static library file for the **ILI9325 LCD** panel (320 x 240 resolution).
- **LCD_16bit_Open407_F407VGT6_9341.lib** – Static library file for the **ILI9341 LCD** panel (320 x 240 resolution).
- **LCD_16bit_Open407_F407VGT6_9481.lib** – Static library file for the **ILI9481 LCD** panel (480 x 320 resolution).
- **LCD_16bit_Open407_F407VGT6_9486.lib** – Static library file for the **ILI9486 LCD** panel (480 x 320 resolution).
- **LCD_16bit_Open407_F407VGT6_8009A.lib** – Static library file for the **OTM8009A LCD** panel (800 x 480 resolution).

Linker and GUI Configuration Steps

1. In the left panel of Keil (**Project Tree**), expand the **KVB_LCD** folder. If a .lib library for another **LCD** panel is already present, remove it: right-click the .lib file and select **Remove File 'xxxx.lib'** from the context menu.
Note: Only one graphic library file must be linked in the project at a time.
2. Right-click the **KVB_LCD** folder in the **Project Tree** and select **Add Existing Files to Group...**
3. In the file selection window, click the **Files of type** drop-down menu at the bottom and switch it to **Library file (*.lib; *.a)**.
4. Select your specific target library file (e.g., **LCD_16bit_Open407_F407VGT6_xxxx.lib**) and click **Add**.
5. Keil MDK is now configured to link this pre-compiled binary file with your main code.
6. Open the main GUI header file **KVB_LCD.h**, locate **line 120**, and uncomment *strictly* the line corresponding to your target **LCD** panel. This line contains the averaged timing configuration for the panel's **FSMC** bus. These values depend heavily on the specific hardware implementation of the panel, wire lengths, and ribbon cable traits.

 **CRITICAL HARDWARE NOTE:** Due to the extremely high data transfer rates into the display's **GRAM**, if you are using a flat ribbon cable, it is **MANDATORY** to install a **22 pF** signal reflection dampening capacitor at the very end of the ribbon cable. Place it as close as physically possible to the panel's interface connector on the **WE (Write Enable)** signal line.

7. Finally, navigate to your project's **main.c** file and uncomment one of the two evaluation functions depending on your display's resolution: **lcd_Test480x320Landscape()** or **lcd_Test320x240Landscape()**.

During the compilation process, **Keil MDK 5** will automatically fetch and link the optimized binary code from your selected library file (e.g., **LCD_16bit_Open407_F407VGT6_9486.lib**).
